

Review

Challenges and Strategies for the Sustainable Environmental Management of Phosphogypsum

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Abstract: Phosphogypsum, a byproduct of phosphate fertilizer production, poses significant environmental challenges due to its large volume, hazardous composition, and radioactivity. Conventional disposal methods, such as stockpiling and landfilling, contribute to soil and water contamination and present risks to human health. This article explores the potential of integrating phosphogypsum into a circular economy framework, focusing on reducing environmental impacts and extracting value from this industrial waste. A detailed assessment of phosphogypsum's chemical composition, including trace metals and radionuclides, is essential for effective management. This review paper examines safe handling, storage, and disposal strategies to minimize environmental risks. Additionally, innovative reuse applications are investigated, such as incorporating phosphogypsum into construction materials like cement, plasterboard, and concrete and its use in agriculture as a soil amendment or for land reclamation. The recovery of critical elements, particularly rare earth elements (REEs), highlights its potential to reduce waste and contribute to meeting the growing demand for strategic resources. Despite its promise, challenges remain, including chemical variability and the presence of radioactive components. This article identifies the technological and regulatory steps required to enable safe, large-scale reuse of phosphogypsum, emphasizing its role in advancing sustainable resource management within a circular economy.



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1. Introduction

Phosphogypsum (PG) waste is produced after phosphoric acid production, a widely adopted method accounting for over 90% of global phosphoric acid manufacturing due to its economic efficiency [1,2]. In this process, phosphate rock, primarily composed of calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$), reacts with concentrated sulfuric acid (H_2SO_4) to produce phosphoric acid (H_3PO_4) and calcium sulfate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), commonly known as gypsum [3]. PG production varies significantly across countries due to differences in the scale of phosphate rock processing industries and industrial practices. PG primarily consists of calcium sulfate dihydrate along with impurities such as fluoride, heavy metals, and naturally occurring radionuclides, including uranium, thorium, and radium based on the source of phosphate rock [3–7]. This process of H_3PO_4 production generates approximately 5 tons of PG per ton of phosphoric acid (P_2O_5), depending on the quality of phosphate rock and process conditions [8–10]. Global PG production exceeds 200 million tons annually, with stockpiles estimated at over 7 billion tons and an annual growth rate of 170 million tons [11,12]. Only about 15% of the PG produced is valorized, while the rest is stockpiled,

posing environmental concerns [1,5]. The global production of PG is substantial, and only a fraction is reused, predominantly in applications such as construction, agriculture, and material recycling, while the majority remains stockpiled [1,3,5,13–15].

Globally, PG stockpiles exceed 830 million tons, with annual increases of 70–80 million tons, yet only 40% is utilized due to barriers in recycling infrastructure and regulatory frameworks [1,16–20]. The total stock of PG stored in tailing ponds between 1961 and 2050 will reach almost 11 billion tons by 2050 [21]. Countries like Brazil, China, Greece, Poland, and the USA face significant PG-related challenges, including water contamination, soil degradation, and public health concerns [1,19,22–24]. For instance, the Wizów Chemical Plant in Poland has over 5 million tons of PG, raising concerns about ecosystem impacts and sustainable reuse potential [10,18,19,22,25,26].

China is the largest producer, generating 22–75 million tons annually [27], followed by the USA, producing 30–50 million tons per year [20,28,29] as shown in Table 1. Other major contributors include Morocco and Russia, each producing around 14–15 million tons annually [30,31], and Tunisia, with 10–12 million tons per year [32,33]. Countries like India and Brazil produce 5–12 million tons and 5.6–10 million tons, respectively. Smaller producers include Poland, Spain, and Vietnam, with estimated outputs of 1.5–3 million tons annually [1,26,34,35]. Croatia and South Korea report 8.5 million and 11 million tons, respectively [15]. The worldwide total PG production ranges between 280 and 300 million tons annually [1]. These figures underscore the substantial global generation of PG, highlighting its significant environmental and industrial implications. Managing this vast quantity requires sustainable approaches to mitigate its impact and promote valorization.

Table 1. Estimated annual PG production for various countries.

Country	Estimated PG Production (Million Tons/Year)	References
Algeria	1.0	[36]
Brazil	5.6–10.0	[25,26,37]
China	22.0–75.0	[26,27]
Croatia	8.5	[15]
India	5.0–12.0	[38]
Jordan	3.0	[14]
Morocco	14.0–15.0	[39,40]
Poland	1.5–2.5	[34,35,41]
Russia	14.0	[31,42]
South Africa	5	[43]
South Korea	11.0	[44]
Spain	2.5–3.0	[16,45]
Syria	0.35	[46]
Netherlands	4.0	[15]
Tunisia	10.0–12.0	[32,42,47,48]
Turkey	3.0	[49,50]
Ukraine	10.0	[35,51]
USA	30.0–50.0	[50,52]
Vietnam	1.2	[53]
Worldwide Total	280–300	[1,25,54]

Phosphogypsum disposal in landfills poses critical environmental and economic challenges due to its radioactive elements, heavy metals, and toxic substances, which leach into soil and groundwater, causing contamination and long-term ecological damage [3,16,19,22,55]. Disposal practices also contribute to drainage blockages, dust pollution, and fluoride release, underscoring the urgency of sustainable solutions [3,56]. Innovative reuse strategies for construction and industrial applications offer promising avenues for

sustainable PG management, provided global collaboration, and stricter regulatory enforcement are achieved [16,19,20,25,57]. Studies have shown that PG's composition varies, and this variability impacts its usability and necessitates localized strategies for processing and application [10,14,15,58]. Advanced characterization techniques, such as X-ray fluorescence, thermogravimetric analysis, and gamma spectroscopy, have identified and quantified PG components, providing insights into their suitability for specific applications [3,59–61].

This review article provides insight into environmental and economic challenges and management associated with phosphogypsum disposal. It addresses critical environmental concerns, including heavy metal leaching and soil, water, and air contamination risks. Traditional management strategies, such as landfilling, surface impoundments, and historical water disposal, are evaluated for effectiveness and associated environmental hazards. In addition to investigating the need for constant future monitoring, sustainable approaches toward PG management can also be utilized. The focus concerns the recycling and repurposing of PG for industrial use like construction, using PG as a soil conditioner, and producing cement, ceramics, and bricks. These measures contribute to the global invention of industrial sustainability, which supports SDG 11 and 12 goals [62].

Moreover, this review article captures the ongoing research on PG attributes and its applications that have been conducted to date, including the work of Outbakat et al. (2022) on PG in agriculture [63]. This study comprehensively examines the generation processes, physical and chemical properties, and potential use of PG as a soil amendment and fertilizer while also addressing its environmental implications. Similarly, research by Pinto et al. (2020) has explored the application of PG in the construction industry, specifically in supersulfated cement mortars [64]. Their findings indicate that PG-based mortars exhibit good durability against sodium sulfate attack but may expand when exposed to magnesium sulfate. Other relevant studies, such as that by Jha et al. (2016), have concentrated on recovering rare earth elements from PG, demonstrating their potential as a resource for valuable elements [65]. These studies highlight the importance of impurity removal and converting PG into high-value products. This work aims to advance the environmental management of PG by mapping existing gaps and proposing innovative approaches for sustainable PG utilization.

1.1. Physical and Chemical Characteristics of PG

Several factors, including the source of phosphate rock, production method, and post-production handling, influence the physical and chemical characteristics of PG, which play a role in its physical and chemical characterization. Phosphogypsum is a fine, powdery material, typically yellowish-white to gray, with a bulk density of 0.9–1.7 g/cm³ and a particle density of 2.27–2.4 g/cm³ [5,20]. Its particle size varies, with 50–75% passing through a 0.075 mm screen, and it contains 10–30% moisture, impacting handling and disposal [5,66]. Initially acidic (pH ~1–2) due to residual acids, its pH gradually neutralizes to 2.5–6 through weathering [5,67]. Over time, weathering and hydration reactions partially convert the phase composition, resulting in changes in crystallinity, pH (which rises to 4–6), and impurity mobility. Phosphogypsum primarily consists of calcium sulfate dihydrate (CaSO₄·2H₂O), which accounts for 85–95% of its mass. However, depending on the production temperature, handling, and storage conditions, minor amounts of calcium sulfate hemihydrate (CaSO₄·0.5H₂O) and anhydrite (CaSO₄) may also be present, especially in aged or weathered samples [5,68]. The properties presented herein reflect typical ranges reported in the literature, encompassing both freshly produced PG and aged material from waste heaps. Understanding these physical properties is crucial for assessing PG's suitability for waste management and various applications.

Phosphogypsum primarily consists of calcium sulfate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), accounting for 85–95% of its composition [3,69]. Other significant components include phosphates (P_2O_5 , 0.41–2.0%) and fluorides (F, 0.57–2.5%), which vary based on the phosphate rock and production process [16,50,67]. Minor elements, such as magnesium (Mg), aluminum (Al), silicon (Si), and iron (Fe), are also present [3,34,55]. PG contains over 50 impurities, including fluorosilicic acid, sodium and potassium fluorosilicates, and undecomposed phosphorus materials, along with heavy metals like cadmium and lead [10,34,70]. Additionally, PG's radioactive elements and organic matter pose environmental and health concerns [69]. Understanding PG's composition is crucial for assessing its environmental impact and potential applications.

Phosphogypsum contains trace metals and radionuclides, including uranium and thorium, which are naturally present in phosphate rock. The concentration of radionuclides, such as Ra-226 and Th-232, varies depending on the geological source of the rock, with regions like Florida (USA) and Morocco having higher levels of radioactivity [15,20,71]. Ra-226, a major contributor to PG's radioactivity, produces radon gas (Rn-222), posing environmental and health risks if not managed properly [8]. Heavy metals, including arsenic, cadmium, lead, and mercury, are also present in PG and can leach into groundwater under specific conditions like low pH and high moisture content [55].

PG is classified as a Technologically Enhanced Naturally Occurring Radioactive Material (TENORM) due to its elevated radionuclide content, particularly Ra-226 and Th-232 [8,72]. Approximately 80% of Ra-226, 30% of Th-232, and 14% of U-238 remain in PG after production [11]. Regulatory bodies, such as Brazil's CNEN, restrict PG use in agriculture or cement production to materials with Ra-226 and Ra-228 levels below 1 Bq g^{-1} [73,74]. REEs, typically 0.04–1% in PG, add economic potential but require careful extraction [10,75]. Radiological hazard indices, such as gamma and alpha indices, are assessed to determine PG's suitability for building materials [76].

1.2. Geographical Variability

The chemical and radiological composition of PG varies significantly based on the geological origin of the phosphate rock. Sedimentary phosphate rock, such as that found in Florida (USA), Morocco, and Tunisia, generally contains higher levels of uranium, thorium, and their decay products, as well as toxic heavy metals like cadmium, arsenic, and lead, resulting in elevated radioactivity and contamination. For instance, Florida's phosphate rock yields PG with some of the highest radionuclide concentrations globally, necessitating additional management safeguards [8,15,29,77].

PG derived from igneous phosphate rock, such as in Brazil, typically has lower concentrations of Cd, As, and mercury but higher levels of barium (Ba). Phosphogypsum from the USA (Idaho) shows high lead content, while PG from Canada is rich in copper, nickel, zirconium, and uranium [78]. Sedimentary-origin PG, with higher impurities and radioactivity, requires stringent handling protocols, while igneous-origin PG may be more suitable for applications like agriculture or construction materials [3,29].

1.3. Disposal Practices

Approximately 58% of PG is dry- or wet-stacked, 28% is discharged into coastal waters, and only 14% is reused, reflecting inefficiencies in global PG management practices [1,3,18,20]. Landfilling PG demands significant financial and spatial resources, with high costs associated with constructing and maintaining secure storage facilities equipped with liners and leachate management systems [3,16]. Ineffective management exacerbates risks, including pollutant release into the environment, soil erosion, and groundwater contamination [19,22,45,55,79,80].

Globally, approximately 85% of phosphogypsum is disposed of in stockpiles near production facilities [20,57]. The estimated global distribution of various phosphogypsum (PG) disposal methods, highlighting the predominance of land-based storage practices such as stacked land, surface impoundments, and engineered landfills, along with less common methods like subsurface and water disposal, is shown in Figure 1 [10,12]. Vast stockpiles exist in the USA, Europe, India, and China, highlighting the need for sustainable management and revalorization of this byproduct [81].

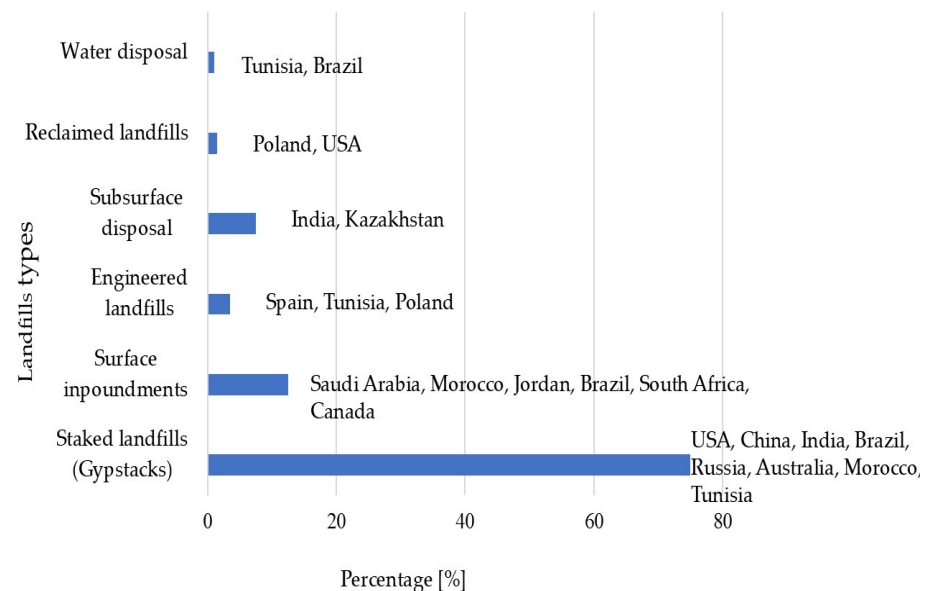


Figure 1. Estimated Global Percentages on Phosphogypsum Disposal Methods: staked land [2,10], surface impoundment [9,67], engineered landfills [82], subsurface disposal [83], reclaimed land [2,84], and water disposal [43,85].

Stacked landfills, commonly known as gypstacks, are the most widely used method for disposing of PG due to their simplicity and cost-effectiveness. Often tens of meters high, these structures store enormous quantities of contaminated material in one location [2,10]. Gypstacks can either be open dumps or engineered landfills. Open dumping involves depositing waste directly on the ground, while engineered landfills include liners and systems to isolate waste from groundwater, reducing environmental risks [86]. Wet stacking is preferred because it can handle storm precipitation but consumes more water. In contrast, dry stacking minimizes water use, making it ideal for arid regions, though it requires regular maintenance to prevent cracking and erosion, which could lead to chemical migration [18]. Poorly managed gypstacks may cause environmental contamination of soil and groundwater through leaching.

Surface impoundments are large basins designed for depositing and settling PG slurry, commonly used in regions with ample land. Examples include Ma'aden Phosphate Company in Saudi Arabia and Morocco's OCP Group, which utilize impoundments near production sites for efficient disposal [10]. While cost-effective, these structures pose environmental risks, such as soil and groundwater contamination from rainwater infiltration and runoff, necessitating careful management to mitigate harm [9,67]. Subsurface disposal involves burying phosphogypsum below ground with impermeable barriers to prevent contamination. It is employed in Rajasthan, India, and Kazakhstan for managing phosphogypsum and mining residues where surface storage is unfeasible [9,10,83,85]. Reclaimed landfills repurpose former phosphogypsum disposal sites for agriculture, industrial use, or recreation after remediation. Examples include Lublin, Poland, where sites are used

for agriculture, and Florida, USA, where gypstacks are transformed into industrial or recreational areas [28,35,42,52].

Water disposal landfills involve discharging PG waste into water bodies, a primarily abandoned method due to environmental concerns. Historically, phosphogypsum was released into the ocean near Santos, Brazil, and dumped into the Mediterranean Sea near Gabès, Tunisia, contributing to heavy metal and radionuclide contamination [9,67]. These waters often contain radionuclide and heavy metal concentrations 3–4 orders of magnitude higher than unperturbed water sources, severely impacting aquatic ecosystems and the food chain, including humans [87]. While some countries like Morocco considered this practice economically viable, its toxicological consequences on marine life, including soluble phosphorus, fluorine, and heavy metals entering aquifers, led to its discontinuation [12]. Sustainable disposal methods now replace water-based practices to minimize environmental damage.

Engineered landfills feature liners, covers, and leachate systems to prevent environmental contamination. Examples include facilities in Tunisia and Huelva, Spain, which manage phosphogypsum safely by controlling runoff, treating effluent, and preventing groundwater seepage and dust emissions. These systems mitigate risks while adhering to strict environmental standards [67]. Table 2 compares phosphogypsum disposal methods, highlighting their advantages and disadvantages. Methods like stacked landfills dominate globally for their simplicity, while engineered and reclaimed landfills prioritize environmental safety and reuse [1,10,20,85]. Surface impoundments and subsurface disposal are region specific [67], and water disposal, though once common, is now nearly obsolete due to environmental concerns [12,85,88].

Table 2. Advantages and disadvantages of different deposits.

Method	Advantages	Disadvantages
Stacked Landfills	• Efficient use of vertical space. • Minimizes land footprint. • It is easier to cap and reclaim for other uses.	• Requires proper engineering to prevent leachate seepage and risk of structural instability • Potential for odor and pest issues if not well-managed. • This can lead to methane emissions if not adequately vented [2,10].
Surface Impoundment	• Cost-effective for liquid waste. • Allows evaporation to reduce waste volume. • It can be adapted to store hazardous materials temporarily.	• High risk of groundwater contamination • Vulnerable to overflow during heavy rainfall or flooding. • Requires extensive monitoring and maintenance.
Engineered Landfills	• Equipped with liners and leachate systems to reduce environmental impact. • Designed to manage diverse types of waste. • Methane capture systems can generate renewable energy.	• High construction and maintenance costs • Limited lifespan; fills up over time. • Requires stringent regulations and monitoring [9,67,86].
Subsurface Disposal	• Effective for managing liquid waste, including hazardous materials. • Minimal land surface impact. • Isolated from direct human contact and ecological exposure.	• Risk of groundwater contamination if systems fail • Long-term monitoring is necessary to prevent leakage. • Expensive installation and maintenance [82,83,88].
Reclaimed Landfills	• Converts waste disposal sites into usable land, such as parks or industrial areas. • Mitigates long-term environmental hazards. • It enhances public perception and reduces the stigma associated with landfills.	• Reclamation can be costly and time-intensive • Potential residual contamination might limit land use. • It requires ongoing monitoring to manage subsurface gases and leachate [3,25,82,85].

Table 2. *Cont.*

Method	Advantages	Disadvantages
Water Disposal Landfills	- Reduces water content in waste, aiding compaction and stability. - Minimizes leachate risks. Less frequent leachate management if systems are appropriately designed. - Capital and operating costs are lower than land-based methods	- Causes toxic metal contamination and turbidity in marine ecosystems - Specialized infrastructure is required for drainage and treatment. - Marine disposal is banned in many countries due to environmental concerns [12,82,88,89].

2. Environmental Impacts and Regulatory Frameworks of PG Waste

Phosphogypsum disposal poses significant environmental challenges due to its hazardous composition, including radioactivity, heavy metals, and toxic elements. Improper management can contaminate air, water, and soil, impacting ecosystems and human health [3,55]. Historically practiced in regions like Tunisia and Brazil, Marine disposal has caused severe ecological disruptions. Landfills, such as gypstacks, contribute to pollution through emissions, leachates, and restricted land use, highlighting the urgent need for sustainable solutions [8,9,26,90]. Sustainable management strategies are essential to mitigate these risks.

2.1. Soil, Water, and Air Contamination

Phosphogypsum can pose significant environmental impacts through soil and water contamination. When improperly managed, its hazardous components, including radionuclides, heavy metals, and fluoride, pose risks to ecosystems and human health. Radium-226 and radium-228, the primary radioactive elements in PG, originate from phosphate rock and persist in the environment. These radionuclides leach from disposal sites into surrounding soil and groundwater, elevating radioactivity levels and creating long-term ecological and health concerns [8,66,90]. Radionuclides accumulate in plants and animals, entering the food chain and increasing bioaccumulation risks. Prolonged exposure to such contamination reduces soil fertility, impacts microbial diversity, and affects ecosystem functionality [30,63,91,92].

Heavy metals, such as arsenic, cadmium, lead, and mercury, further exacerbate soil pollution. These metals, commonly found in PG, leach into the soil over time, contaminating groundwater and surrounding environments [3,15,92–96]. Heavy metal contamination disrupts microbial activity, hinders biochemical processes, and lowers soil fertility. It also risks human and animal health through the food chain as plants absorb these metals [97,98]. Soil acidification, often caused by heavy metal contamination, reduces pH levels, impacting vegetation and soil microbial communities [99].

Fluoride contamination in PG compounds causes soil degradation. Fluoride exists as hydrofluoric acid and other compounds, which leach into the soil, affecting its physical properties and nutrient availability. High fluoride concentrations disrupt plant growth, reducing photosynthesis, causing chlorosis, and inhibiting root elongation [26,100]. In regions like Florida and Tunisia, where PG is deposited in large quantities, fluoride contamination has significantly degraded soil, reduced biodiversity, and affected agricultural productivity [101,102].

Recent research has highlighted the importance of understanding the chemical behavior of impurities in PG for effective environmental management and valorization. As detailed by Ennaciri and Bettach et al. [40], major impurities, including fluoride, phosphate, heavy metals (e.g., As, Cd, Pb), and radionuclides, exhibit varying degrees of mobility and environmental persistence, depending on factors such as pH, redox conditions, and mineral associations. The study emphasizes that certain impurities are more likely to leach under

acidic conditions, posing risks to soil and water systems. Therefore, tailoring management practices and valorization pathways to the specific impurity profiles of PG is crucial for minimizing environmental risks and optimizing recovery processes. The composition of PG varies across geographical sources, influencing contamination levels. PG from Tunisia and Morocco has higher radium concentrations, while PG from the USA contains more lead and cadmium [20,79,92,93,103]. This variability necessitates region-specific disposal and management strategies to mitigate environmental risks [26,88]. Natural processes, such as rainfall, worsen contamination by dissolving PG components, producing leachates that contaminate groundwater and surface waters. PG leachates often contain radionuclides and heavy metals at concentrations 3–4 orders of magnitude higher than unperturbed waters. These contaminants make water unsafe for consumption and disrupt aquatic ecosystems when they reach rivers and lakes [26,104].

Exposure to weathering processes enables the leaching of hazardous elements, causing long-term soil acidification and reduced fertility [66]. Additionally, radionuclides like polonium-210 and radon from PG piles migrate into groundwater, increasing environmental risks [105].

2.2. Health and Ecological Risks

Phosphogypsum also contributes significantly to airborne pollution, posing health risks and long-term environmental challenges. Poorly managed PG waste piles release particulate matter containing radioactive radon gas and toxic heavy metals such as arsenic, cadmium, lead, and mercury [106–108]. Radon, a decay product of radium in PG, is a significant concern due to its carcinogenic effects, particularly lung cancer, especially among smokers and populations near disposal sites [20,88,109]. Airborne dust from PG piles can exacerbate respiratory conditions like asthma and bronchitis. Chronic exposure to heavy metals in this dust may result in kidney damage, neurological disorders, and cancer [25,82,110]. Furthermore, airborne pollutants from PG often settle on crops and soils, leading to the bioaccumulation of harmful substances in the food chain and the threat to food safety [8]. Landfills and PG stacks also emit greenhouse gases (GHGs) like methane, carbon dioxide, and nitrogen oxides through decomposition and chemical reactions, contributing negatively and positively to global warming. Methane emissions, in particular, have a high global warming potential and adversely affect atmospheric quality. PG does not emit methane directly, but when co-disposed with organic waste, it alters microbial activity, suppressing methanogens and promoting sulfate reducers, thereby significantly reducing methane (CH₄) emissions, a potent greenhouse gas with a global warming potential that is 25 times greater than CO₂. This mitigation effect makes PG a contributor to climate change mitigation by helping reduce overall greenhouse gas emissions in agricultural and composting systems [111,112]. Sulfur dioxide and nitrogen dioxide emissions from landfills also harm the environment and can cause respiratory infections and irritations for nearby populations [113,114]. Table 3 summarizes the potential health effects associated with common contaminants found in phosphogypsum, including heavy metals, radionuclides, and fluoride, as well as their exposure pathways.

Studies worldwide, including those in South Africa, Syria, and Brazil, have shown that communities near PG disposal sites experience increased respiratory and health issues due to airborne contaminants [115,116]. For instance, 78% of people living near South African landfills reported health effects linked to air pollution, emphasizing the severity of this issue [117]. Several strategies are recommended to mitigate these risks. Covering PG stacks with protective barriers or using wetting agents can reduce dust dispersion [76]. Advanced dust suppression systems and encapsulation techniques can further minimize airborne particulate release, while impermeable layers can reduce radon emissions [118].

Regular air quality monitoring near PG sites is essential to identify and address pollution hotspots [119,120]. Airborne pollution from phosphogypsum disposal sites poses serious environmental and health risks. Releasing radon, heavy metals, and GHGs contributes to respiratory diseases, cancer, and global warming. PG stack storage can contribute indirectly to global warming when exposed to environmental conditions that trigger chemical and physical transformations. Over time, PG stacks may release volatile sulfur compounds, such as hydrogen sulfide (H_2S), under moist and anaerobic conditions, particularly in the presence of sulfate-reducing bacteria. Improper disposal management may lead to land degradation and increased CO_2 emissions due to loss of carbon sinks [121,122]. Weathering of PG can lead to the release of calcium and sulfate ions into soil and water systems, which may react with nitrogen compounds to enhance the emission of the greenhouse gas nitrous oxide (N_2O), with nearly 300 times the global warming potential of CO_2 . Conversely, beneficial uses, such as composting or mineral carbonation, have shown potential to reduce CH_4 and N_2O emissions, thereby supporting climate change mitigation efforts [123,124]. Large-scale PG disposal sites may also cause land use changes, such as deforestation and vegetation loss, reducing carbon sink capacity and increasing net CO_2 emissions [122]. PG stacks over time have the cumulative emissions and land degradation effects associated with long-term PG storage, which must be considered as indirect contributors to global warming. Implementing mitigation measures, such as improved site management, dust control technologies, and air quality monitoring, is essential to minimize these impacts and safeguard public health and the environment [5,20,116]. Table 3 summarizes the health effects of common contaminants found in phosphogypsum, categorized by exposure pathways (airborne, water, and crops).

Table 3. Summary of Health effects of common contaminants found in phosphogypsum.

Contaminant	Exposure Pathway	Health Effects	References
Radium-226	Airborne (dust)	Increased risk of bone cancer, leukemia, and anemia due to radioactive decay products (radon gas).	USEPA 1999: Radiation at Superfund Sites [125] IAEA, 2013 [5] WHO2011: Guidelines for Drinking-water Quality, 4th Edition [126] UNSCEAR, 2021: Report on Sources and Effects of Ionizing Radiation [7]
	Water (leaching)	Elevated risk of internal radiation exposure, kidney damage, and soft tissue cancers.	
Fluoride	Water	Dental and skeletal fluorosis with chronic exposure; neurological effects in high concentrations.	WHO 2011: Guidelines for Drinking-water Quality, 4th Edition [126] ATSDR 2012: Toxicological Profile for Cadmium [127]
Cadmium	Crops, Water	Kidney damage, skeletal demineralization, and increased risk of cancer through bioaccumulation.	WHO 2010: Guidelines for Drinking-water Quality, 4th Edition [126] Codex Alimentarius Commission, CXS 193-1995 [128]
Arsenic	Water, Crops	Increased risk of skin cancer, cardiovascular disease, and neurological toxicity.	WHO 2010: Guidelines for Drinking-water Quality, 4th Edition [126] Codex Alimentarius Commission: (CXS 193-1995) [128]
Lead	Airborne, Crops	Neurological impairment in children, kidney damage, and hypertension in adults.	USEPA 2006: Radiation at Superfund Sites [125] WHO 2011: Guidelines for Drinking-water Quality, 4th Edition [126] Codex Alimentarius Commission, CXS 193-1995 [128]
Uranium	Water, Crops	Nephrotoxicity, chemical toxicity in kidneys, and increased risk of cancer due to radiation.	ATSDR 2013: Toxicological Profile for Uranium [129] IAEA 2013: Safety Standards for the Management of NORM Residues [5]

Table 3. *Cont.*

Contaminant	Exposure Pathway	Health Effects	References
Chromium	Water, Crops	Chromium VI exposure can cause lung cancer, kidney damage, and skin irritation.	ATSDR 2012: Toxicological Profile for Chromium [130] WHO 2010: Guidelines for Drinking-water Quality, 4th Edition [126]
Thorium	Airborne	Increased risk of lung and pancreatic cancers due to radioactive decay.	USEPA 2021: Radiation at Superfund Sites [125] WHO 2011: Guidelines for Drinking-water Quality, 4th Edition [126]
Zinc	Water, Crops	It can disrupt gastrointestinal function and immune response with chronic overexposure.	ATSDR 2005: Toxicological Profile for Zinc [131]
Nickel	Airborne, Water	Allergic reactions, respiratory issues, and increased cancer risk in high exposures.	ATSDR 2007: Toxicological Profile for Nickel [132]
Barium	Water	It can lead to gastrointestinal symptoms, muscle weakness, and cardiovascular toxicity.	ATSDR 2007: Toxicological Profile for Barium [133]

2.3. Economic Consequences

Disposing of PG entails significant economic consequences for industries and governments. Currently, 34 landfills use PG as a filling aggregate, while addressing the demand for mine-filling materials comes with considerable economic burdens. The costs of processing, transportation, storage, and compliance with stringent environmental regulations ensuring environmental safety significantly outweigh the potential benefits [3,134]. Costs associated with landfill management, leachate control, and monitoring to mitigate risks, particularly from radioactive elements, impose heavy financial burdens on industries, especially in regions with strict disposal policies [66,67]. Improperly managed disposal sites devalue land, contaminate groundwater, and necessitate expensive remediation, compounding financial pressures on industries and communities [3]. Moreover, environmental contamination from PG stacks risks public health crises, leading to costly litigation, compensation for affected populations, and reputational damage to industries [20]. These costs are amplified by the need for containment systems to mitigate leachate, dust, and gas emissions and by implementing monitoring systems to comply with environmental regulations and remediation of contaminated sites further compounds economic burdens [24,87,135]. The leaching of toxic heavy metals, radionuclides, and fluoride from PG piles into soil and groundwater often necessitates costly cleanup operations to restore environmental health. For instance, groundwater contamination by PG has been linked to significant financial outlays for water treatment and remediation projects, especially in regions where PG disposal sites are poorly managed [117]. Additionally, the degradation of soil fertility due to PG-related acidification imposes financial burdens on agricultural sectors, requiring expensive soil amendments to restore productivity [96]. Airborne pollution from PG disposal sites also incurs economic costs. Dust emissions and hazardous gases, such as radon and sulfur dioxide, necessitate investments in air quality management, including dust suppression systems and protective coverings for waste piles [76]. Poor air quality caused by PG waste has been linked to public health issues, increasing healthcare expenses for respiratory diseases, cancer treatments, and other pollution-related illnesses [9,88,135]. These health-related costs disproportionately affect communities near PG disposal sites and can strain public health systems or communities requiring compensation from manufacturing plants. Climate change impacts further add to the economic consequences of PG waste. Landfills and PG stacks contribute to greenhouse gas (GHG) emissions, including methane and carbon dioxide, which exacerbate global warming. The financial toll of climate change, such as damage from extreme weather events, rising sea levels, and disruptions to agriculture, indirectly links PG disposal to broader economic challenges [45,136].

Several manufacturing plants have been closed due to the high costs of complying with PG disposal regulations. For example, the Piney Point phosphate plant in Florida, USA, shut down in 2001 after struggling with costly environmental management for its gypstack, including wastewater containment upgrades [8]. Similarly, the Wizów phosphogypsum plant in Poland ceased operations as the financial burden of meeting stricter environmental standards, such as safe storage and groundwater protection, became unsustainable [10]. In Tunisia, several phosphate plants also faced closure due to the rising costs of engineered landfills and waste remediation efforts [67]. Investments in advanced processing technologies and regulatory incentives for PG reuse could alleviate these economic burdens while driving sustainable development and market competitiveness [137,138].

2.4. Regulatory Frameworks and Challenges

2.4.1. International Guidelines on Phosphogypsum Management

Several applicable international guidelines have been established to address the management of PG due to its environmental, health, and economic implications. These frameworks emphasize sustainable management, resource recovery, and safe disposal practices.

1. The International Atomic Energy Agency (IAEA) has developed detailed guidelines under the Safety Reports Series No. 78, focusing on managing Naturally Occurring Radioactive Materials (NORM) such as PG. The IAEA outlines protocols for radiological characterization, controlled storage, worker safety, and disposal. For example, PG stacks must include impermeable liners and leachate collection systems to prevent groundwater contamination, while reuse criteria ensure PG with low radioactivity is safely repurposed [5].
2. The United Nations Environment Programme (UNEP) advocates a circular economy approach through its Sustainable Use of Industrial By-products framework. UNEP emphasizes resource recovery, such as extracting REEs, and promotes alternative uses of PG in construction and agriculture. It also highlights the importance of environmental risk assessments and policy support to encourage innovation and commercialization of PG-based products [139].
3. The Food and Agriculture Organization (FAO) provides specific guidelines for PG use in agriculture, detailed in the FAO Soils Bulletin No. 62. These include strict thresholds for impurities like cadmium and lead to prevent soil and crop contamination. Application rates depend on soil and crop types, with safeguards to avoid areas with shallow groundwater. Farmer training on safe storage, handling, and application is also emphasized to balance agricultural benefits with environmental protection [140].
4. The US Environmental Protection Agency (EPA) regulates PG management under frameworks like the Environmental Protection Authority Act (EPA)-1968. The EPA quantified ocean-dumping practices, including 4.5 million tons of industrial waste, and has established restrictions to mitigate PG's environmental risks [134,141].

2.4.2. European Union Guidelines on Phosphogypsum Management

The European Union (EU) has established comprehensive guidelines for managing phosphogypsum within its industrial waste and radiation protection regulatory framework. These guidelines aim to mitigate environmental and health risks while promoting the sustainable utilization of PG.

1. The European Commission's Radiological Protection Principles Concerning the Natural Radioactivity of Building Materials (RP-112, 1999) provides a foundation for PG management in the EU. These principles require radiological characterization of PG to ensure that its radioactivity levels, particularly from radium-226, thorium-232, and uranium-238, meet safety thresholds. The guidelines recommend limiting the gamma

dose rate from building materials made with PG to a maximum of 1 millisievert per year [142].

2. PG is classified as a waste material under the European Waste Framework Directive (2008/98/EC), which mandates proper handling to prevent environmental contamination. The directive emphasizes impermeable liners and leachate collection systems in PG storage facilities to prevent groundwater pollution. It also provides criteria for PG reuse, ensuring compliance with safety standards to minimize health and environmental risks. Member states are encouraged to develop strategies for reducing PG waste through recycling and reuse [143].
3. The EU Circular Economy Action Plan (2020) aligns with the Sustainable Development Goals (SDGs) by promoting resource recovery, such as the extraction of REEs from PG. It advocates alternative uses for PG in construction materials (e.g., cement and plasterboard) and agriculture as a soil conditioner. These practices aim to reduce reliance on virgin resources and foster a sustainable waste management system [144].
4. EU legislation mandates comprehensive environmental risk assessments to mitigate PG's potential impacts. For instance, the Water Framework Directive (2000/60/EC) requires member states to monitor and manage the leachability of heavy metals and radionuclides from PG to protect water resources. Similarly, the Industrial Emissions Directive (2010/75/EU) sets strict emission limits for industries managing PG, requiring leachate control systems, emission monitoring, and advanced pollution abatement technologies [145,146].
5. The Industrial Emissions Directive (2010/75/EU) specifically addresses phosphoric acid plants that produce PG, requiring the adoption of the Best Available Techniques (BAT) to reduce emissions and environmental impacts. It focuses on controlling dust, heavy metals, and radionuclides during PG production and storage, implementing impermeable liners for leachate containment, and ensuring regular reporting of emissions data to ensure compliance [145].
6. The European Green Deal emphasizes a low-carbon, resource-efficient economy by 2050. It supports the reuse of PG in agriculture and construction, provided safety and environmental standards are met. The EU provides funding for research into innovative PG applications, such as carbon sequestration in construction and soil improvement techniques [62,147].

2.4.3. National Regulation of PG Management Practices by Countries in the EU

1. Poland

In Poland, PG management is subject to stringent regulations under the Polish Environmental Protection Law and aligned with EU directives, including the Waste Framework Directive (2008/98/EC). These regulations mandate safe storage and disposal of PG, emphasizing minimizing environmental contamination through impermeable liners and leachate collection systems. While efforts to reuse PG in construction have been explored, concerns about radioactivity and heavy metal content, as regulated by the Basic Safety Standards Directive (2013/59/EURATOM), have limited large-scale adoption. Current research focuses on extracting REEs and other valuable components, aligning with the EU Circular Economy Action Plan and Poland's resource efficiency goals [10,148,149].

2. Spain

Spain's management of PG is governed by national regulations implementing the Waste Framework Directive (2008/98/EC) and the Industrial Emissions Directive (2010/75/EU). In Huelva, one of Europe's largest PG storage sites, strict environmental controls are enforced to rehabilitate stacks through vegetation and stabilization

techniques. Additionally, PG is regulated for reuse in cement production and soil remediation, provided it meets safety thresholds outlined in the Radiological Protection Principles (RP-112, 1999). Spain also complies with the Water Framework Directive (2000/60/EC) to monitor and control leachability of pollutants from PG storage sites, ensuring protection of water resources [142,143,145,146].

3. Belgium

In Belgium, PG management adheres to EU regulations, including the Waste Framework Directive (2008/98/EC) and the Basic Safety Standards Directive (2013/59/EURATOM). The Wallonia region has implemented advanced processing techniques to reduce PG's environmental impact, focusing on reuse in road construction and as a sulfur source for agricultural fertilizers. The Industrial Emissions Directive (2010/75/EU) plays a key role in regulating emissions during PG handling and ensuring that reuse applications comply with safety standards for radiation and environmental protection [143,145,148,150].

4. Croatia

Croatia manages PG under EU directives, particularly the Waste Framework Directive (2008/98/EC) and the Basic Safety Standards Directive (2013/59/EURATOM). In regions like Slavonia, PG is being researched for potential reuse in agriculture and construction. Concerns over radioactivity and heavy metal content limit large-scale adoption, but recent efforts focus on stabilizing PG stacks and exploring its application in producing environmentally friendly building materials [15,143,148].

5. Netherlands

The Netherlands adopts a proactive approach to PG management, emphasizing sustainability and circular economy principles under the Waste Framework Directive (2008/98/EC). Advanced technologies are used to extract sulfur and REEs from PG, promoting resource recovery. The Industrial Emissions Directive (2010/75/EU) ensures tightly controlled emissions during PG processing and storage. Additionally, PG is being repurposed in road construction and as a filler in concrete, with safety ensured by the Basic Safety Standards Directive (2013/59/EURATOM) and national environmental regulations [143,145,148].

2.5. Gaps in Regulation and Monitoring Standards

2.5.1. Inconsistencies in Radiological Standards Across Countries

A significant gap in PG management policies is the inconsistent radiological safety standards across countries. For instance, radon emission thresholds differ widely, which is critical for assessing environmental and health impacts. The United States enforces strict radon limits for PG stacks under the Clean Air Act, while many developing nations lack clear or enforceable guidelines [82]. Such disparities create challenges for industries operating across borders, complicating compliance efforts and fostering unequal environmental protections. Inconsistent standards also hinder global management of NORM, as economic priorities may outweigh environmental safeguards in less regulated regions [5]. Harmonizing global frameworks, such as those advocated by the International Atomic Energy Agency (IAEA Safety Reports Series No. 78) [5], is essential to bridge these disparities and ensure comprehensive and equitable PG management. Additionally, pollution control efforts are increasingly addressing inter-media impacts, such as acidification of lakes and soil contamination from PG disposal [60].

2.5.2. Lack of Harmonized Global Protocols for Long-Term Stack Monitoring

The absence of universally accepted protocols for long-term monitoring of phosphogypsum stacks represents a critical gap in global management practices. While countries

like the United States mandate periodic environmental assessments and structural integrity checks under frameworks like the Resource Conservation and Recovery Act (RCRA), these measures are not standardized worldwide. This lack of consistency leaves aging stacks vulnerable to environmental hazards, including leachate seepage, structural collapses, and increased radon emissions [82,151]. Effective long-term monitoring protocols should mandate continuous radon measurement, groundwater quality testing, and structural stability assessments to mitigate such risks. The International Atomic Energy Agency (IAEA Safety Reports Series No. 78) highlights the importance of ongoing monitoring to manage NORM, but its adoption remains uneven across regions [5]. Global guidelines could address these gaps, ensuring consistent application of best practices to safeguard human and environmental health. Harmonized standards would reduce variability in management approaches, particularly in countries with limited regulatory frameworks [25,150].

2.5.3. Gaps in EU Policy on Secondary Uses of PG in Non-Agricultural Sectors

The EU has made progress in PG management through regulations like the Waste Framework Directive (2008/98/EC) and the Industrial Emissions Directive (2010/75/EU) [143,145]. However, gaps remain in promoting secondary uses of PG outside the agricultural sector. Current EU policies primarily address environmental risk mitigation, offering limited support for innovative applications such as using PG in construction materials, road stabilization, or as a source of sulfur in industrial processes [152]. This lack of emphasis on secondary uses undermines the circular economy objectives of the European Green Deal, which aim to maximize resource recovery and reduce waste. Despite the potential to reuse PG in sectors like construction, concerns over heavy metals and radioactivity have stalled large-scale adoption due to the absence of clear, harmonized safety standards [149]. Addressing these gaps requires targeted research funding to explore viable PG reuse technologies and develop industry-specific guidelines for non-agricultural applications. Financial incentives and policy support could encourage industries to invest in innovative processing methods while ensuring compliance with safety thresholds outlined in the Basic Safety Standards Directive (2013/59/EURATOM) [148]. Clear safety standards and robust risk assessments would enable sustainable reuse while mitigating environmental and health risks.

2.5.4. Need for Integrated Policies Addressing Environmental Risks and Economic Opportunities

Current policies treat PG as a waste product, strongly emphasizing mitigating environmental risks such as radionuclide emissions and heavy metal contamination. While these measures are crucial, they often neglect PG's significant economic potential, including its value as a source of critical raw materials like REEs and its potential for secondary uses in construction, road stabilization, and industrial processes [2,57].

Integrated policies are needed to maximize the dual benefits of environmental protection and economic opportunity. Such policies should balance resource recovery with risk mitigation by incorporating frameworks for extracting valuable materials, providing incentives for adopting sustainable technologies and encouraging cross-sector collaboration to develop innovative PG applications [147]. The European Green Deal and Circular Economy Action Plan emphasize the importance of turning industrial byproducts like PG into valuable resources. However, the practical implementation of these strategies remains limited by regulatory gaps and fragmented policy approaches. Integrated policies would create a unified decision-making framework that addresses environmental pressures while fostering economic growth and societal sustainability [147,149].

2.5.5. Policy Challenges for Reuse

Despite growing recognition of PG's potential for reuse in various industries, significant policy challenges hinder its widespread adoption. One major obstacle is the absence of harmonized global standards for determining the safety of PG reuse in applications such as construction, road stabilization, and industrial processes. Countries implement varying thresholds for contaminants like radium-226 and heavy metals, complicating international trade and industrial collaboration [5,150]. In the European Union, policies such as the Basic Safety Standards Directive (2013/59/EURATOM) and the Waste Framework Directive (2008/98/EC) provide a framework for PG management [143,148]. However, gaps remain in promoting its reuse beyond agricultural applications. For instance, industries face challenges meeting stringent radiological and environmental safety standards, often requiring expensive testing and processing technologies. Additionally, a lack of financial incentives and research funding limits innovation in developing cost-effective reuse applications, particularly in lower-income regions where industries prioritize economic feasibility over environmental sustainability [62]. Public perception and regulatory hesitancy are another challenge. Concerns about the potential health risks associated with PG reuse, such as exposure to radioactive elements and toxic heavy metals, deter its integration into mainstream industrial practices. Without clear, evidence-based guidelines and effective communication strategies, industries struggle to gain regulatory and societal acceptance for PG reuse [45]. To address these challenges, policies must focus on creating uniform safety standards, providing financial and technical support for innovation, and establishing clear industry guidelines. Integrating reuse policies into broader circular economy strategies can help align environmental goals with economic opportunities, transforming PG into a resource rather than a liability.

3. Sustainable Management and Applications of PG

3.1. Waste Minimization

Minimizing PG waste and ensuring its safe storage is fundamental to addressing the environmental, economic, and regulatory challenges associated with its disposal. As a by-product of the phosphate fertilizer industry, PG presents significant risks due to its radioactivity, heavy metal content, and large volume, necessitating comprehensive waste reduction and containment strategies. Reducing PG generation begins with optimizing industrial processes to lower impurity levels. Improvements in sulfuric acid utilization and refining reaction conditions in wet-process phosphoric acid production can produce cleaner PG with fewer contaminants, facilitating its reuse [5]. Integrating closed-loop systems in fertilizer plants is another practical approach. These systems recycle process water and recover residual materials, minimizing waste streams and aligning with circular economy principles [57].

Despite these advancements, global PG utilization remains low, with only 40% recycled. Impurities such as radionuclides, including radium, are significant barriers to reuse. Studies suggest that PG should not be classified solely as waste but as a resource, highlighting its potential in construction materials, agriculture, and industrial applications [16,57]. For example, Croatia has explored recycling PG for road stabilization and construction, while other regions focus on recovering REEs to enhance resource efficiency [19]. The waste management hierarchy prioritizes waste prevention and recycling, with disposal as a last resort, as shown in Figure 2. Emphasizing valorization strategies, such as using PG in soil stabilization or as an asset in Portland cement production, reduces environmental risks and storage costs while supporting global sustainability goals.

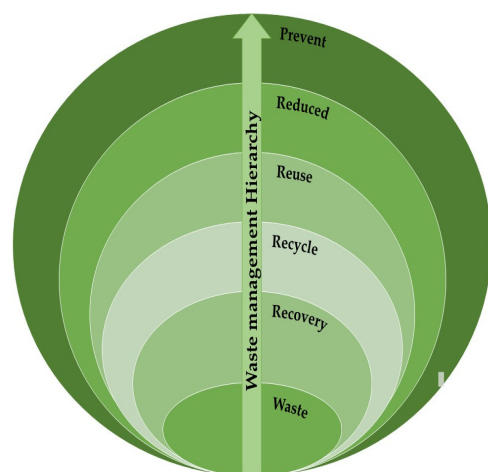


Figure 2. The hierarchy of waste management.

3.2. Safe Storage

Safe storage is critical to prevent environmental contamination for significant volumes of PG that cannot yet be reused. Often located near production sites, PG stacks must adhere to stringent engineering standards. Impermeable liners and leachate collection systems are essential to prevent seepage into groundwater, while effective drainage systems manage surface water and mitigate erosion risks [134]. Radon emissions from PG stacks pose a significant environmental and health concern. Installing radon barriers and using advanced monitoring systems, such as real-time radon and leachate detectors, can prevent environmental disasters and ensure compliance with safety standards [153]. Regular environmental assessments, including groundwater quality checks and structural integrity inspections, are critical for maintaining stack safety over time. Green cover systems are an innovative solution to stabilize PG stacks. Vegetative covers reduce erosion, suppress dust emissions, and enhance soil stabilization. They also provide a buffer against adverse weather conditions, contributing to the overall environmental safety of storage sites [19]. Some countries have implemented sustainable practices, such as engineered landfills, to safely store PG while minimizing environmental risks. For example, in Huelva, Spain, over 1200 hectares of PG are stabilized using vegetation and advanced containment techniques [85,117]. Similarly, South Korea and China have safely adopted measures to contain PG, reflecting the global shift toward sustainable waste management practices [44]. Minimizing PG waste generation and ensuring its safe storage is critical for mitigating environmental impacts and unlocking its potential as a valuable resource. Adopting cleaner production methods, closed-loop systems, and advanced storage technologies can address these challenges while promoting economic and environmental sustainability [57,78].

3.3. Technological Innovations for Environmental Protection

Technological innovations play a pivotal role in mitigating the environmental impact of PG. One such advancement is the chemical treatment of PG, such as neutralization with lime or other alkaline substances. This process stabilizes hazardous elements like heavy metals and radionuclides, significantly reducing their mobility and environmental risks and making PG safer for storage or reuse [1]. The integration of IoT-enabled sensors and predictive maintenance systems has revolutionized stack monitoring. These technologies provide real-time data on parameters such as radon emissions, stack stability, and leachate quality, enabling proactive measures to prevent environmental hazards. Advanced landfills equipped with liner systems, gas extraction setups, and leachate monitoring fur-

ther enhance containment and resource recovery, including converting landfill gas into energy [134].

Research on PG has also focused on improving its quality and expanding its applications. Purification techniques and recovery of valuable components, such as REEs, are gaining traction. These methods reduce impurities and create new avenues for utilizing PG in construction, agriculture, and industrial applications [19].

Global sustainability efforts emphasize life-cycle approaches to PG management, addressing environmental risks from production to disposal and promoting systemic solutions. Innovations in landfill engineering, such as anaerobic and aerobic bioreactors, further improve waste decomposition and energy recovery, highlighting the potential of modern technologies in PG management [62].

3.4. Applications and Reuse Potential

Phosphogypsum has more than 50 identified potential applications, and the most applicable sectors are agriculture, construction materials, landfills, and road construction, which rank as the top sectors by volume, shown in Figure 3. However, only about 15% of global PG production is currently recycled, with the rest remaining unused due to challenges such as impurities and radioactivity [2,14].

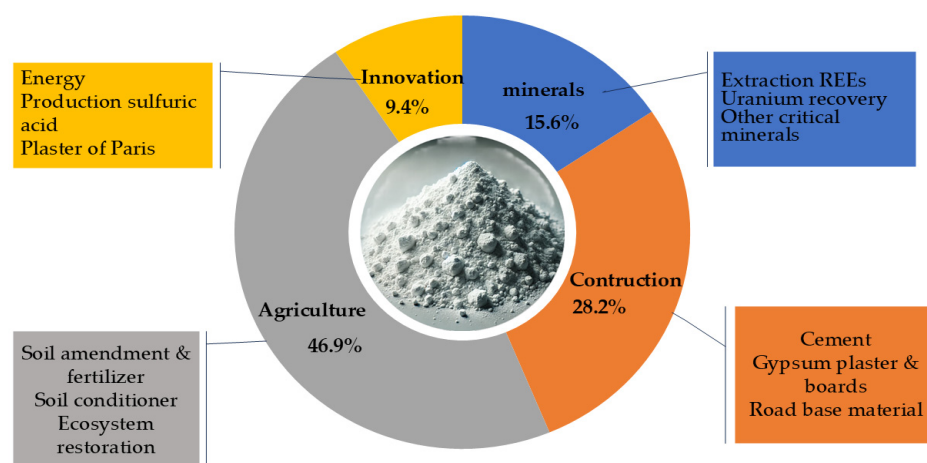


Figure 3. Applications of phosphogypsum.

3.4.1. Construction Materials (Cement, Concrete, Boards, Roads)

Phosphogypsum has garnered increasing attention as an alternative material in construction, offering opportunities to reduce waste, conserve natural resources, and lower costs. Its cement, concrete, boards, and road applications demonstrate its potential to support sustainable practices while addressing environmental concerns associated with PG disposal.

PG is widely researched as a substitute for natural gypsum in cement production. PG reduces dependency on mined gypsum as a setting regulator, thus contributing to environmental sustainability. PG can be thermally converted into a gypsum binder, commonly known as hemihydrate gypsum ($\text{CaSO}_4 \cdot 0.5\text{H}_2\text{O}$), through calcination at temperatures ranging from 130 to 180 °C. This thermally treated PG serves as a binder widely used in construction applications and prefabricated products. Several studies have demonstrated that PG, when properly purified and processed, can yield comparable mechanical properties to natural gypsum binders, including compressive strengths exceeding 8 MPa and adjustable setting times through the addition of modifiers [154,155]. Moreover, PG can replace up to 100% of natural gypsum raw materials, depending on the degree of pretreatment and impurity removal [156]. The technical feasibility of producing PG-based

binders has been demonstrated in both laboratory and pilot-scale experiments [157], and the environmental benefits of such valorization strategies are well recognized, particularly in terms of reducing storage burdens and secondary pollution [84]. Studies by Leonid Dvorkin et al. [51] and Bumanis et al. [158] confirmed that PG can replace natural gypsum in Portland cement without affecting setting time or mechanical properties, reducing energy consumption during production. Degirmenci et al. [49] reported that PG improves early strength and durability when added to concrete. Additionally, Calderón-Morales et al. [159] highlighted the potential of PG in geopolymer composites, a sustainable alternative to traditional cement binders. However, impurities such as phosphate and fluoride require advanced purification techniques before PG can be used in large-scale applications [3,12,159]. The incorporation of PG into concrete mixes enhances compressive strength and stability. Studies by Anouar et al. [160], Li et al. [161] and Ahmed et al. [162] demonstrated that PG-modified concrete shows improved mechanical performance, making it suitable for diverse construction applications. Concrete products incorporating PG offer several advantages. PG enhances workability, reduces water demand, and can partially replace cement or natural gypsum, thereby lowering both production costs and environmental impact. Studies have shown that PG-modified concrete can maintain compressive strengths above 20 MPa while reducing CO₂ emissions by up to 15% due to the reduction in clinker usage [163,164]. Additionally, PG improves setting time control and sulfate resistance, which makes it suitable for various structural and non-structural applications. Research on geopolymer systems combining PG with Supplementary Materials, such as fly ash, further supports their application in pavement materials and high-strength concrete [165,166].

Phosphogypsum's chemical properties make it an excellent substitute for natural gypsum in producing plaster and plasterboards. Research by Haneklaus et al. [62] and Levickaya et al. [156] showed that PG can produce high-quality gypsum plaster with enhanced fire resistance and thermal insulation. Its use reduces the carbon footprint of production by minimizing energy consumption compared to traditional gypsum [57,62]. Du et al. [167] demonstrated that PG-based gypsum boards exhibit improved mechanical strength and water resistance, making them suitable for interior and exterior applications. Guan et al. [50] corroborated these findings, emphasizing the durability of PG-based boards in humid environments. Japan, France, and Germany are leaders in developing PG-based plasterboard technologies, highlighting the material's viability [168].

Studies by US-EPA [134] and Mehta et al. [169] indicated that PG improves road-based materials' load-bearing capacity and durability. PG-stabilized soils exhibit reduced plasticity, facilitating easier handling and compaction during construction. This review highlighted that PG-stabilized soils exhibit improved physical and strength performance, including plasticity index reductions and compaction characteristics enhancements. Research by Tomašević et al. [170] demonstrated that PG is environmentally viable for road construction, with minimal risks of groundwater contamination and radiological impacts. A report on phosphogypsum for secondary road construction exhibited adequate strength and durability, performing well in service with negligible environmental risks [171]. Morocco's road construction industry utilizes PG as an embankment aggregate, reducing transportation costs and minimizing environmental damage. Stabilized PG blends with lime or fly ash also serve as effective road-based materials, as confirmed by research published in Applied Sciences, which highlighted that PG positively influences the strength development of sediment-PG mixtures, making them suitable for subgrade material in road construction [30,172].

3.4.2. Agricultural and Land Reclamation Uses of PG

PG has increasingly been utilized in agricultural and land reclamation applications due to its rich composition in calcium, sulfur, phosphorus, and trace elements like zinc (Zn) and silicon (Si), which makes it a valuable material for enhancing soil fertility, improving crop yields, and rehabilitating degraded lands.

Phosphogypsum is widely used as a soil amendment and fertilizer additive due to its ability to improve soil structure and fertility. Calcium strengthens plant cell walls and supports root development, while sulfur enhances photosynthesis, protein synthesis, and nutrient uptake. These nutrients are vital for improving crop quality and yield [19,173]. PG is particularly effective in neutralizing soil acidity, making it beneficial for crops grown in acidic soils, such as rice, maize, and wheat. It improves water retention, reduces soil crusting, and increases phosphorus availability, critical for root development and fruit production [13]. Gharaibeh et al. [174] demonstrated PG's effectiveness in reducing salinity and improving fertility in saline soils. PG is also a sustainable alternative to traditional phosphorus fertilizers, addressing concerns about the rapid depletion of natural phosphorite resources. Recycling PG reduces dependency on mineral fertilizers, promoting sustainable agriculture [13,57,175]. Gabsi et al. [176] showed that combining PG with organic amendments improved germination rates, soil fertility, and crop yields in degraded soils. Trace elements in PG, such as zinc and silicon, further enhance its agricultural benefits. Zinc is critical for seed development, protein synthesis, and grain ripening, while silicon strengthens plant structures, enhances stress tolerance, and improves resistance to pests and diseases [1,177].

Phosphogypsum is a proven soil conditioner for reclaiming degraded soils affected by salinity, alkalinity, or acidification. It replaces sodium in sodic soils with calcium, improving soil structure, permeability, and water infiltration. Gharaibeh et al. [174] reported a 90% reduction in sodium content in sodic soils treated with PG, demonstrating its effectiveness in soil reclamation. PG reduces aluminum toxicity in acidic soils, a common barrier to plant growth, by neutralizing excess acidity and improving soil pH. It enhances soil aggregation, reduces erosion, and improves moisture retention, contributing to long-term soil health [178]. Recent studies have highlighted additional agricultural applications of PG beyond soil amendment. PG has been effectively used in the development of mineral-based slow-release fertilizers, offering a sustainable solution for nutrient delivery while utilizing industrial waste [164]. Furthermore, PG-derived materials have demonstrated promising performance in the remediation of farmland fluoride pollution, offering a cost-effective and efficient method for immobilizing fluoride and enhancing soil quality in affected regions [179].

PG has shown great potential in reforestation and ecological restoration projects. Enhancing soil fertility and providing essential nutrients for root development supports native vegetation growth in degraded lands. PG has also been used to rehabilitate oil-contaminated lands. Research on the complex processing of phosphogypsum suggests that PG can improve soil structure and functionality, enabling vegetation regrowth and ecological restoration [31]. The study indicates that PG application can enhance soil aeration and microbial activity, accelerating the breakdown of hydrocarbons in contaminated soils and, thus, facilitating ecosystem recovery.

A comprehensive review by Silva et al. [19] discussed various applications of PG, emphasizing its effectiveness in enhancing soil structure and fertility and promoting sustainable vegetation growth. The study highlights that PG applications can lead to improved water retention, reduced soil erosion, and the restoration of degraded lands. PG has also been used to rehabilitate oil-contaminated lands. Various research has been performing that suggests PG can improve soil structure and functionality, enabling vegetation regrowth

and ecological restoration. The study indicates that PG application enhances soil aeration and microbial activity, accelerating the breakdown of hydrocarbons in contaminated soils and facilitating ecosystem recovery [1,180].

3.4.3. Recovery of Valuable Elements

The recovery of REEs from phosphogypsum has gained significant interest due to their critical role in advanced technologies, renewable energy systems, and national defense. Phosphogypsum contains trace amounts of REEs, including lanthanum, cerium, and neodymium, which can be recovered using methods such as acid leaching with hydrochloric acid (HCl) or nitric acid (HNO₃) and solvent extraction [75]. These elements are essential for manufacturing permanent magnets, energy storage devices, and catalysts, making their recovery from PG an important area of research. While the REE content in PG is relatively low, typically ranging from 0.3% to 0.6%, the vast quantities of phosphogypsum produced annually make it a potential secondary resource [10]. According to Ramirez et al. (2021), up to 70% of REEs from phosphate ores end up in phosphogypsum during the production of phosphoric acid, highlighting its recovery potential [181]. China currently dominates REE production, controlling over 68.7% of the global supply and driving the search for alternative sources [182]. Research has focused on improving extraction technologies, such as selective leaching and advanced solvent extraction methods [183–186]. The recovery of REEs from PG is being increasingly explored as a cost-effective and environmentally sustainable alternative to conventional mining methods. Traditional REE mining—typically from ores such as bastnäsite, monazite, or xenotime—involves extensive excavation, chemical beneficiation, and waste generation, often resulting in environmental degradation, high energy consumption, and significant capital investment. PG is a waste byproduct of phosphate fertilizer production that already contains appreciable quantities of REEs in a chemically accessible form, making it an attractive secondary source. Compared to conventional mining, recovering REEs from phosphogypsum presents both economic and environmental advantages. While traditional REE mining involves high capital costs and significant environmental degradation, studies suggest that REE recovery from PG may cost around USD 15–25 per kg of mixed rare earth concentrate, compared to USD 30–70 per kg from primary ores, depending on grade and processing route. Moreover, utilizing PG reduces waste stockpiles and offsets disposal costs, making it a potentially cost-effective and sustainable alternative [187,188]. Despite the progress, economic and environmental challenges remain in scaling these technologies for industrial use. Further studies have been conducted on the techno-economic assessment of REE recovery from phosphoric acid sludge—a related byproduct—which suggest that integrating REE recovery with phosphoric acid production can be profitable under certain conditions. For example, the study presents scenarios where the net present value (NPV) over 12 years ranges from USD 178.7 million to USD 441.8 million, depending on the specific recovery process implemented. While this study focuses on sludge rather than PG, it highlights the broader economic viability of REE recovery in phosphate-based industrial systems. It reinforces the importance of further optimizing PG recovery processes [184].

Phosphogypsum also contains trace amounts of uranium, a valuable resource for nuclear energy production. Uranium recovery from PG is typically achieved through acid leaching, solvent extraction, and ion exchange [66]. During phosphoric acid production, approximately 90% of the uranium in phosphate rock is transferred to phosphoric acid, with about 10% remaining in phosphogypsum [5]. The recovery of uranium from PG addresses waste management concerns and provides a sustainable source of this critical material. Studies have demonstrated promising extraction efficiencies, with advancements in process optimization and solvent extraction improving feasibility [189,190]. Kiegiel et al. [186]

explored the recovery of uranium and other valuable metals from substrates and wastes in the copper and phosphate industries, including phosphogypsum. The study highlighted the potential for integrating uranium recovery into industrial waste management practices, utilizing efficient separation techniques such as liquid–liquid extraction and advanced membrane systems. This approach recovers uranium and extracts other valuable metals, making it economically viable and environmentally sustainable. Innovations in the recovery process have led to improved extraction efficiencies and reduced environmental impacts. For example, IAEA [191] and Dong et al. [192] have evaluated the kinetics and economics of uranium leaching from phosphogypsum, finding that process optimization can significantly lower operational costs. However, challenges remain in managing radioactive materials and ensuring compliance with safety regulations [66].

In addition to REEs and uranium, phosphogypsum contains various other critical minerals and heavy metals that can be recovered. These include copper, zinc, vanadium, molybdenum, chromium, cobalt, nickel, and thorium. Recovering these elements from PG mitigates environmental concerns and provides an alternative source of valuable resources for various industrial applications. Copper and zinc are among the heavy metals that can be recovered from phosphogypsum. Copper is widely used in electrical and electronic industries due to its excellent conductivity, while zinc is essential for galvanization and alloy production. Efficient recovery of these metals can be achieved through acid leaching and solvent extraction methods, as demonstrated in studies exploring the potential of phosphogypsum as a secondary resource for industrial metals [193]. Vanadium can be obtained from PG using alkali leaching, and solvent extraction techniques have shown promise in recovering these metals. Huang et al. [194] reported the successful recovery of vanadium and molybdenum using innovative hydrometallurgical methods, emphasizing the economic viability of such processes. Chromium, nickel, and cobalt are often present in trace amounts in phosphogypsum and can be extracted using advanced leaching techniques followed by ion exchange or solvent extraction [195]. The recovery of these elements aligns with circular economy principles and reduces dependency on traditional mining. Chen et al. [196] demonstrated the feasibility of recovering tantalum using hydrometallurgical methods, highlighting their potential for industrial-scale operations.

3.4.4. Environmental and Industrial Uses of Phosphogypsum

Phosphogypsum, a byproduct of phosphate fertilizer production, offers various environmental applications that contribute to sustainable waste management and environmental protection. Among its key uses are mine backfilling, wastewater treatment, and as a landfill liner material.

- **Mine Backfilling**

Phosphogypsum has been explored as a backfill material for underground mines to fill voids created during mining operations. This application offers several environmental benefits, including reduced surface land disturbance, stabilization of mine structures, and prevention of ground subsidence. Its gypsum content provides mechanical stability, while its chemical properties support mine stabilization [197–199]. However, the potential for leaching trace elements and radioactivity necessitates strict compliance with environmental safety standards to ensure safe implementation [55].

- **Wastewater Treatment**

Phosphogypsum has shown also potential in wastewater treatment due to its high calcium content, which enables it to neutralize acidic wastewater. Additionally, phosphogypsum can adsorb heavy metals, such as lead, cadmium, and arsenic, from industrial effluents, reducing contamination levels in wastewater. This eco-friendly and cost-effective

application utilizes a waste byproduct for environmental remediation. However, monitoring the leaching potential of harmful substances is critical to prevent secondary pollution [176,200–203].

- Landfill Liner Material

Moreover, phosphogypsum has been evaluated as a material for landfill liners due to its low permeability and ability to form stable, impermeable layers. As a barrier material, it helps prevent the leaching of hazardous substances into the environment, providing an effective waste containment solution [19,25]. Despite its potential, environmental concerns related to radioactivity and the potential release of trace elements must be addressed to ensure its long-term safety [5,9,14].

- Sulfuric Acid Production

Phosphogypsum is rich in calcium sulfate, making it a potential raw material for sulfuric acid production. Sulfuric acid is a vital industrial chemical in mineral processing, battery manufacturing, and petroleum refining. By processing phosphogypsum, sulfur dioxide (SO_2) can be released and converted into sulfuric acid through oxidation and hydration processes [36,204]. This approach not only recycles waste but also mitigates environmental hazards associated with phosphogypsum disposal. However, challenges such as managing SO_2 emissions and ensuring the safe handling of phosphogypsum require strict compliance with environmental standards [205].

- Building Paints

Phosphogypsum is utilized in the production of building paints as a filler and extender. Its fine particle size and high calcium sulfate content enhance the texture, durability, and opacity of paints. Additionally, using phosphogypsum as an additive reduces production costs by replacing more expensive materials. Incorporating phosphogypsum in paints also offers an eco-friendly solution by repurposing waste. However, concerns regarding trace element leaching and product safety must be addressed [1,57].

- Plaster of Paris (POP)

Phosphogypsum serves as a raw material for producing Plaster of Paris (POP), a widely used construction material known for its quick-setting properties and smooth finish. The process involves heating phosphogypsum to produce calcium sulfate hemihydrate, the primary component of POP [206]. Using phosphogypsum in POP production reduces waste disposal issues and provides a sustainable alternative to conventional raw materials. However, regulatory compliance is essential to manage its radioactivity and ensure safe application in construction [207].

- Energy Production

The sulfur content in phosphogypsum can be extracted for energy production. The recovery of sulfur from phosphogypsum involves converting calcium sulfate into sulfur dioxide (SO_2), which is further processed into sulfuric acid. This conversion occurs through a series of chemical reactions, including the thermal decomposition of CaSO_4 to release SO_2 . The exothermic nature of these reactions generates substantial heat, which can be harnessed for energy production [190,204,208]. This process not only recovers sulfur for industrial use but also provides a sustainable energy source by utilizing the heat generated during the reaction. The recovered sulfur can be used in industries such as fertilizers, chemicals, and petroleum refining, offering both economic and environmental benefits [193].

Thermal treatment methods, including calcination and pyrolysis, can convert phosphogypsum into lime (CaO) and sulfur dioxide. These processes involve heating phosphogypsum to high temperatures (typically above 600°C), causing it to decompose and release SO_2

gas. Potentially, the heat energy produced during these reactions could be used to power turbines or provide thermal energy for industrial applications [136]. Calcination produces lime as a byproduct, which has multiple industrial applications, including construction and wastewater treatment. This dual benefit of material recovery and energy generation makes thermal treatment an efficient approach to recycling phosphogypsum [166,209].

- Carbon Capture

Phosphogypsum has significant potential for use in carbon capture technologies, particularly for reducing greenhouse gas emissions from industrial processes. As a byproduct rich in calcium sulfate (CaSO_4), PG can undergo chemical modifications to enhance its CO_2 adsorption capabilities. Modified PG materials have been developed to increase their reactivity with carbon dioxide, allowing for efficient mineral carbonation. This process converts CO_2 into stable carbonate compounds, providing a long-term storage solution for the captured gas [45,210]. One of the main methods of using phosphogypsum for carbon capture is mineral carbonation, in which the calcium content of phosphogypsum reacts with CO_2 to form calcium carbonate. This not only sequesters CO_2 but also produces valuable by-products, such as precipitated calcium carbonate, which can be used in construction and manufacturing [200]. Phosphogypsum has demonstrated potential for CO_2 sequestration through mineral carbonation, particularly by converting to calcium carbonate (CaCO_3). Studies estimate that 1 ton of PG can sequester approximately 0.2–0.3 tons of CO_2 , depending on purity and reaction conditions [43,104]. Integration with CCUS technologies—such as flue gas CO_2 capture—can enhance environmental benefits, turning PG valorization into a dual-purpose pathway for both waste management and carbon mitigation. Innovations in CO_2 sequestration with phosphogypsum include its integration with membrane electrolysis, enhanced carbonation processes under elevated pressures, and the development of composite materials with enhanced adsorption properties [51]. By combining phosphogypsum with other industrial wastes, researchers have also improved its efficiency as a carbon sink, making it a viable option for large-scale CO_2 capture projects [45]. Life cycle assessment (LCA) studies have demonstrated that substituting natural gypsum with phosphogypsum (PG) in industrial applications, such as cement and plaster production, can significantly reduce environmental impacts. For example, replacing 1 ton of natural gypsum with PG can result in a carbon footprint reduction of 150–300 kg CO_2 -eq, primarily due to the avoidance of mining and transportation emissions [99,158,185]. These results highlight the environmental benefits of PG valorization and support its integration into circular economy models [163].

- Decorative Tiles

Phosphogypsum has been successfully utilized in the manufacture of decorative tiles, providing a sustainable and cost-effective alternative to traditional building materials. Decorative tiles made from phosphogypsum are lightweight and exhibit excellent thermal insulation and soundproofing properties. Their natural texture and color can be enhanced with pigments and surface finishes, making them suitable for both interior and exterior applications [211,212]. The production process typically involves blending phosphogypsum with binders like Portland cement, lime, or polymers, followed by moulding and curing. Additives can be incorporated to improve water resistance, durability, and aesthetic appeal. These tiles are versatile and can be customized for various architectural styles, reducing reliance on non-renewable resources. One of the key benefits of using phosphogypsum in tile production is its low environmental impact. It not only diverts waste from disposal sites but also reduces the energy consumption associated with producing conventional ceramic tiles [213]. Furthermore, the inclusion of phosphogypsum enhances the mechanical

strength and thermal stability of the tiles, making them suitable for high-performance applications [2,19].

- Production of Ammonium Sulfate and Calcium Carbonate

Recent studies have demonstrated that PG can be effectively converted into valuable industrial products through wet chemical processes. These transformations include the synthesis of calcium carbonate (CaCO_3), calcium hydroxide (Ca(OH)_2), ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$), and sodium sulfate (Na_2SO_4), offering a sustainable route for PG management. Carbonation techniques allow for the precipitation of CaCO_3 and Ca(OH)_2 from PG under controlled conditions [43]. Similarly, PG can undergo a two-step ammonia-carbonation reaction to yield ammonium sulfate and calcium carbonate, with process parameters, such as the $\text{OH}^-/\text{Ca}^{2+}$ molar ratio, critically influencing the conversion efficiency. The synthesized products, $(\text{NH}_4)_2\text{SO}_4$ and CaCO_3 , are highly recommended for use in both the fertilizer and cement industries without any reservations [103]. These methods offer the dual benefit of environmental remediation and resource recovery. Moreover, studies have proposed procedures for scaling up these conversion methods to industrial levels, highlighting their potential economic and environmental impact [35,45].

- Functional Filler in Composite Materials

PG has emerged as a promising functional filler in polymeric, cementitious, and metal composites, offering both performance and environmental benefits. Recent advancements in polymer science have demonstrated that surface-modified polyglycerol can enhance its tensile strength, thermal resistance, and flame retardancy, thereby improving its compatibility with matrices such as polypropylene and epoxy resins [213,214]. Hydrophobically modified PG used in polypropylene composites resulted in a $\uparrow 12\%$ increase in tensile strength and improved water resistance, making it suitable for construction and outdoor applications [215]. In another study, Zhang et al. [210] incorporated up to 60% polyglycol in polymer composites designed for pipelines, achieving flexural strengths exceeding 20 MPa and excellent dimensional stability. Further research conducted by Dong et al. [201] evaluated PG-based fillers in asphalt mixtures, confirming their mechanical compatibility and potential for harmless treatment. Other research demonstrated that PG improves the compressive strength and hardness of aluminum-based systems, attributed to the refinement of microstructures and the formation of stable intermetallic compounds. These diverse applications underscore PG's growing utility in high-performance materials across construction, automotive, and industrial sectors, transforming hazardous waste into a valuable resource aligned with circular economy principles [35,196].

- Oil–Water Separation Applications

Recent advances have demonstrated the potential of PG in oil–water separation, through surface modification. Due to the PG porous structure, high surface area, and calcium sulfate composition, it serves as a suitable substrate for hydrophobic functionalization. Researchers have developed hydrophobically modified PG-based membranes and filters by incorporating silanes, graphene oxide, or other surface-active agents [155]. These materials exhibit excellent water repellence and oil selectivity, with research demonstrating separation efficiencies over 95%, even after multiple reuse cycles. These systems are promising for treating oily wastewater from industries such as petrochemical, textile, and food processing [216]. Moreover, hydrotalcite and polymer-based foams offer a relevant comparison for high-efficiency separation and oil adsorption, demonstrating similar functionality in terms of selective wetting behavior and material reusability. The modification process is relatively low-cost, making it attractive for scalable industrial applications [155,216].

3.5. Barriers to Reuse and Circular Economy Integration

The reuse of phosphogypsum is promising for industrial, agricultural, and environmental applications. However, significant challenges related to technological barriers, economic feasibility, and environmental safety, particularly concerning radioactive contaminants, impede its widespread adoption. This section reviews the primary challenges in phosphogypsum reuse.

3.5.1. Technological Barriers

- Processing Techniques

Phosphogypsum, a byproduct of phosphate fertilizer production, varies in chemical composition based on its source. Transforming this material into usable products requires specialized processing techniques, including grinding, calcination, and chemical modification [217]. These processes aim to enhance its physical and chemical properties to meet the requirements of industrial or agricultural applications [48]. Developing cost-effective and scalable processes for phosphogypsum reuse is one of the most critical challenges. Phosphogypsum requires processing to meet the quality standards of various applications, such as construction materials, soil amendments, and industrial raw materials. However, these techniques are often energy-intensive and may involve complex operational steps. Thermal treatment, such as calcination, consumes large amounts of energy, making it less cost-effective for large-scale applications. Chemical modification processes, including the addition of binders or stabilizers, can improve phosphogypsum's functionality but often result in higher production costs. The scalability of these processes presents another major challenge. While laboratory-scale experiments demonstrate promising results, achieving similar efficiency and cost-effectiveness in industrial-scale operations remains difficult [218]. Advanced technologies, such as bioleaching and plasma treatment, show potential for improving processing efficiency, but they are still in the developmental stage and require further optimization for cost-effective implementation [203].

- Separation of Contaminants

Phosphogypsum contains impurities such as heavy metals, radioactive elements, and organic residues, which limit its applications. Efficient separation of these contaminants is essential for ensuring their safety and suitability for reuse. Techniques such as acid leaching, ion exchange, and membrane filtration have been employed to remove contaminants [219,220]. Despite their effectiveness, these techniques are often expensive and energy-intensive, making them unsuitable for large-scale applications. Current methods, such as leaching and ion exchange, are expensive and difficult to scale up for industrial applications [175]. For example, leaching processes require significant amounts of reagents, generating secondary waste streams that need further treatment [221]. Similarly, ion exchange and membrane technologies, though efficient, involve high operational costs and require frequent maintenance. Developing innovative separation methods that are both cost-effective and scalable is crucial. Emerging techniques, such as bioleaching and advanced filtration systems, show promise for improving contaminant removal efficiency while minimizing environmental impacts [222].

3.5.2. Economic Feasibility and Market Acceptance

Recycling PG faces economic challenges, including high processing costs and limited market acceptance. The cost-effectiveness of recycling phosphogypsum depends on the availability of cost-efficient processing technologies and the market demand for its derived products [12,57]. The competition from traditional materials further complicates the economic feasibility of phosphogypsum reuse. For instance, conventional materi-

als like natural gypsum or lime are often cheaper and more readily available, making phosphogypsum-derived products less competitive [9,57].

Additionally, the lack of financial incentives and subsidies for PG recycling discourages industrial investment. Markets are dominated by traditional materials, consumers may be reluctant to switch to phosphogypsum-derived products without clear evidence of their safety, durability, and cost-effectiveness [88,208]. Market acceptance also hinges on addressing safety concerns regarding phosphogypsum's potential environmental and health risks. Public awareness of the environmental and economic benefits of phosphogypsum reuse remains low. Education campaigns and pilot projects showcasing successful applications can play a key role in shifting public perception [2,62,66]. Public perception of its association with radioactivity and heavy metals can limit its use in construction and agriculture. Educating stakeholders and implementing stringent safety measures can help build confidence in phosphogypsum products [28,82].

3.5.3. Managing Radioactive Elements

One of the most significant challenges in PG reuse is its radioactive element content, particularly radium-226 and uranium isotopes. These radionuclides, originating from phosphate ores, pose environmental and health risks if not properly managed [72]. Regulatory standards on radioactivity levels in materials used for construction and agriculture are stringent in many countries. For example, the permissible levels of radium in building materials often limit phosphogypsum's direct use. Compliance with these standards requires effective monitoring and remediation techniques [5,25,74]. The leaching of radioactive elements from PG during its application is another major concern. Studies have highlighted the risk of radium and uranium leaching into soil and groundwater, necessitating the development of stabilization techniques to immobilize these contaminants [223,224]. Innovative approaches, such as encapsulation, thermal treatment, and chemical stabilization, have been explored to mitigate the risks associated with radioactive contaminants. Additionally, advanced monitoring systems and regular environmental assessments are essential to ensure the safe use of phosphogypsum [75,225].

3.5.4. Circular Economy for PG

Phosphogypsum offers significant potential for contributing to a circular economy through sustainable practices. This material can be transformed into a valuable resource, minimizing its environmental impact and creating economic benefits. Advancing a circular economy for phosphogypsum presents significant opportunities to transform this by-product into a valuable resource. By applying principles of resource recovery, recycling, and extended product lifecycles, industries can achieve sustainable waste management while fostering economic growth. Integrating phosphogypsum into industrial processes, coupled with innovative reuse strategies, supports the transition to a more sustainable and circular economy. Additionally, the underutilization of PG represents a missed economic opportunity. PG contains valuable components such as sulfur, calcium, and REEs, which could be recovered for economic gain. However, limited infrastructure and technological investment hinder its processing and reuse, preventing contributions to circular economy principles, job creation, and environmental waste reduction [62,149,226].

The application of circular economy principles to phosphogypsum not only addresses waste management but also supports industrial ecology by integrating waste reuse into economic systems. The circular economy emphasizes reducing waste and maximizing resource efficiency by keeping materials in use for as long as possible. For phosphogypsum, this involves reprocessing, recycling, and repurposing it into valuable products. Integrating phosphogypsum into industrial processes is a cornerstone of circular economy strategies.

By utilizing phosphogypsum in diverse applications, industries can reduce waste, lower costs, and decrease reliance on virgin resources [12,147].

3.6. Case Studies: Phosphogypsum Recycling

The recycling of PG has gained global attention as industries and governments recognize its potential to address resource scarcity and environmental challenges. Several successful case studies demonstrate innovative applications of phosphogypsum across various sectors, highlighting its economic and environmental benefits.

In Europe, the EU Gypsum Gap Initiative addresses a significant shortfall in gypsum supply caused by the Renewable Energy Directive's restrictions on coal-fired power plants. This shortage, particularly evident in Germany, has prompted the integration of phosphogypsum as a substitute for flue gas desulfurization (FGD) gypsum in construction materials. Pilot projects have demonstrated that phosphogypsum can effectively replace natural gypsum in cement and plasterboard production without compromising product quality. For example, in Germany, studies revealed that phosphogypsum-based drywall exhibits comparable compressive strength and durability to natural gypsum, while in France, phosphogypsum was successfully used in producing plasterboard, reducing reliance on virgin gypsum sources [62].

In the fertilizer industry in Spain, a pilot project focused on recovering REEs from phosphogypsum using a two-step leaching process with organic acids. This innovative approach successfully extracted high-purity REEs such as neodymium and cerium, showcasing the economic feasibility of REE recovery from phosphogypsum while addressing waste management issues [2]. The project demonstrated the potential to integrate waste management and resource extraction, reducing environmental risks associated with phosphogypsum storage. Similar efforts are being pursued in China, where REE recovery from phosphogypsum is being driven by the demand for strategic materials in renewable energy technologies [227].

In India, a cement manufacturer developed a phosphogypsum-based concrete blend that reduced production costs by 20% while improving performance in infrastructure projects. The blend demonstrated enhanced compressive strength and durability, particularly in high-humidity environments, making it an attractive option for large-scale infrastructure development. This case highlights the scalability of using phosphogypsum in construction, contributing to both cost savings and environmental sustainability [228].

In Tunisia, phosphogypsum was used in agriculture to improve soil fertility and crop yields. Combined with organic manure, phosphogypsum was applied to degraded soils over two years, resulting in reduced soil salinity and acidity, improved water retention, and nutrient availability. Crop yields increased by 15–30%, depending on the crop type. This study underscores the potential of phosphogypsum as a sustainable soil conditioner, particularly in arid regions where soil degradation is a significant challenge [176].

In China, PG was utilized for sulfuric acid production, a process that involved the thermal decomposition of calcium sulfate to release sulfur dioxide, which was then converted into sulfuric acid. This method achieved an 80% sulfur recovery efficiency and significantly reduced phosphogypsum waste, while the heat generated during the process was used to power plant operations. This case demonstrates the feasibility of integrating phosphogypsum recycling into chemical manufacturing, aligning with circular economy principles [229].

Phosphogypsum has also been explored for carbon capture through mineral carbonation. A European project demonstrated that phosphogypsum could react with CO₂ under controlled conditions to achieve high sequestration rates, with up to 90% of CO₂ emissions captured in experimental setups. The resulting byproducts were stable and suitable for

use in construction, showcasing the dual benefits of waste recycling and climate change mitigation [200,230].

These case studies illustrate the diverse applications of phosphogypsum in construction, agriculture, energy production, and environmental management. They highlight the material's potential to address critical resource and environmental challenges while supporting the transition to a circular economy. By scaling up these innovations, industries and governments can transform phosphogypsum from a waste byproduct into a valuable resource [2,19,225].

4. Future Perspectives and Recommendations

As global demand for sustainable practices grows, phosphogypsum reuse and recycling present significant opportunities. However, challenges related to technical feasibility, environmental safety, and policy frameworks must be addressed. This section explores future directions in material recycling, monitoring, and regulatory advancements while identifying critical research gaps and proposing actionable solutions.

4.1. Innovations in Material Recycling and Resource Recovery

Advances in recycling technologies and resource recovery methods are essential for maximizing the potential of phosphogypsum in diverse applications. Innovations in extraction technologies, such as solvent extraction, ion exchange, and bioleaching, are enabling the recovery of valuable elements like rare earth elements and uranium with greater efficiency while incorporating green chemistry principles to minimize waste and environmental impact [15,19,20,231]. Additionally, advanced reprocessing methods, including thermal treatments like calcination and chemical modifications such as alkaline activation, are transforming phosphogypsum into high-value products like construction materials and fertilizers while reducing emissions [69]. Synergistic recycling approaches, which combine phosphogypsum with other industrial byproducts like fly ash or slag, further optimize material recovery by reducing waste generation and enhancing the functionality of end products for various industrial applications [232].

4.2. Enhanced Monitoring and Characterization Techniques

Improved monitoring and characterization techniques are essential for understanding phosphogypsum's properties and ensuring its safe application across various industries. Radiological monitoring plays a critical role, with sensitive methods designed to detect radioactive isotopes such as radium and uranium, ensuring compliance with safety standards. The integration of real-time monitoring systems using IoT-enabled sensors further enhances safety during storage and application by providing continuous data and early warnings [223]. Advanced spectroscopy and imaging techniques, including X-ray diffraction (XRD), scanning electron microscopy (SEM), and inductively coupled plasma mass spectrometry (ICP-MS), offer detailed insights into the mineralogical and chemical composition of phosphogypsum. These tools are invaluable for determining its suitability in applications such as construction materials or fertilizers. Furthermore, automated real-time monitoring systems, equipped with machine learning algorithms, can track changes in phosphogypsum properties during processing and storage, ensuring operational efficiency and optimizing safety protocols [233].

4.3. Policy Recommendations

Strong policy frameworks are critical for ensuring the safe and sustainable reuse of phosphogypsum across various sectors. Establishing harmonized global standards for permissible levels of radioactive and chemical contaminants is a priority to address safety concerns while enabling widespread adoption in agriculture, construction, and

other industries. These regulations must balance environmental safety with the practical use of phosphogypsum [223]. Financial incentives, including subsidies, tax credits, and grants, can play a pivotal role in encouraging industries to invest in recycling technologies. Additionally, government funding for research and development initiatives can expand phosphogypsum applications and enhance its economic viability [62]. Multidisciplinary collaboration between researchers, industry professionals, policymakers, and environmental organizations is essential for addressing the technical, regulatory, and economic barriers to phosphogypsum reuse. Such partnerships can foster innovative solutions and create comprehensive strategies for safe and effective utilization [5,141]. Public awareness and education campaigns are equally important for improving societal acceptance. By highlighting the safety and environmental benefits of phosphogypsum recycling through transparent communication of safety assessments and pilot project outcomes, policymakers can build trust and promote the integration of phosphogypsum-derived products into mainstream markets [57,193].

4.4. Emerging Technologies and Future Recommendations

Future innovations and strategies should prioritize key areas to transform phosphogypsum from a challenging byproduct into a valuable resource, contributing to a sustainable and circular economy [62]. Integrating rare earth element extraction technologies into existing phosphoric acid plants is crucial for maximizing resource recovery and creating economic incentives for industries. This approach leverages existing infrastructure and supports the recovery of critical materials needed for high-tech applications. Deploying IoT-based sensors for real-time monitoring of radon emissions, stack stability, and ground-water quality can significantly enhance safety and operational efficiency during storage and application. These advanced systems ensure compliance with environmental standards while optimizing resource use [3,9,14,234].

Phosphogypsum also holds the potential for carbon sequestration, aligning with global climate goals by offering a dual benefit of waste recycling and greenhouse gas reduction. Utilizing it in carbon capture and storage technologies can contribute to sustainable industrial practices while mitigating environmental impacts [45]. Furthermore, harmonizing global policy is essential for establishing consistent international radiological and chemical safety standards. These standards would ensure uniformity in phosphogypsum classification, reuse, and disposal practices, facilitating global trade and recycling initiatives [222]. By addressing these priorities, phosphogypsum can be repurposed effectively, reducing its environmental footprint while creating new opportunities for industries and advancing sustainability goals [12,116,149].

4.5. Research and Policy Directions

Strategic Pathways for Collaboration

Collaboration is a cornerstone for overcoming the multifaceted challenges associated with phosphogypsum reuse and unlocking its full potential across industries. Strategic pathways for collaboration focus on fostering partnerships among governments, industries, academic institutions, and international organizations to promote innovation, streamline regulatory frameworks, and enhance public trust. By leveraging these collaborative pathways, stakeholders can address the technical, regulatory, and economic barriers to phosphogypsum reuse. Coordinated efforts will enable sustainable practices, advance circular economy goals, and transform phosphogypsum from an environmental challenge into a valuable resource. These pathways include the following:

1. **Public–Private Partnerships (PPPs):** Governments and industries can establish PPPs to co-fund research and development (R&D) initiatives to advance phosphogypsum

recycling technologies. These partnerships can pool resources to scale up innovative solutions, such as extracting rare earth elements or converting phosphogypsum into sustainable construction materials [141].

2. **International Collaboration:** International cooperation is essential given the global nature of phosphogypsum production and its environmental impacts. Collaborative efforts can focus on harmonizing safety standards, sharing best practices, and developing transboundary agreements for safe reuse [62,223].
3. **Academic–Industry Partnerships:** Academic institutions and industries should work together to bridge knowledge gaps and accelerate the development of advanced technologies for phosphogypsum recycling. Joint research projects can focus on innovative applications, such as carbon capture, wastewater treatment, and advanced construction materials, aligning with circular economy principles [1,57,70].
4. **Stakeholder Engagement:** Engaging local communities, environmental organizations, and policymakers is crucial for building public trust and ensuring socially responsible practices. Transparent communication about the safety and benefits of phosphogypsum reuse can increase market acceptance and address public concerns about radioactivity and environmental risks [5,204].
5. **Global Knowledge Networks:** Establishing platforms for knowledge sharing among researchers, industries, and policymakers can drive innovation and reduce duplication of efforts. Conferences, workshops, and online repositories can facilitate the exchange of data, case studies, and research findings, fostering a collaborative environment for sustainable phosphogypsum management [235].
6. **Funding and Incentives:** Governments and international bodies should create funding mechanisms and provide financial incentives to encourage collaboration on phosphogypsum projects. Subsidies for research, tax breaks for industries adopting recycling technologies, and grants for pilot projects can drive participation and innovation [2,3,15,20,62].

5. Conclusions

This review article highlights the multifaceted potential of phosphogypsum management in various applications, demonstrating its value as a raw material for industrial, agricultural, environmental, and energy recovery solutions. Different research has shown the potential of phosphogypsum, which will play a vital role in environment management and land conservation. In conclusion, various innovative approaches can be applied to overcome the challenges associated with phosphogypsum management and maximize its sustainable use, and a multifaceted approach is necessary:

- Investing in advanced technologies for the extraction of REE with higher efficiency recovery and safe contaminant removal;
- Industrial integration incorporating phosphogypsum as raw material in sectors will enhance resource efficiency and support circular economy goals by offering sustainable alternatives and decreasing reliance on non-renewable materials in industrial processes;
- Regulatory frameworks and policymakers should establish harmonized global standards to ensure safe phosphogypsum reuse, address radiological concerns, and promote trade through financial incentives and awareness campaigns to enhance market acceptance;
- Public Awareness of the reuse of phosphogypsum encouraging its trade through financial incentives and awareness campaigns will help build market acceptance;

- Collaboration among industries, researchers, policymakers, and environmental organizations is essential to share best practices and overcome technical, regulatory, and market challenges through public–private partnerships;
- Academia should close knowledge gaps, improve the current industrial processes, and develop novel applications like carbon sequestration to expand sustainable phosphogypsum reuse;
- Global Cooperation and International collaboration are essential to standardizing practices, addressing radioactive contamination, and fostering knowledge exchange. This ensures consistent progress across regions.

By prioritizing these strategies, phosphogypsum can transition from an environmental burden to a valuable resource, contributing to sustainable development goals. Innovation, policy support, and interdisciplinary collaboration can unlock its full potential while minimizing environmental impacts and maximizing economic benefits.

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